

Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

Video generation • World models • 4D scene understanding

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Research interests span video understanding and generation, multimodal foundation models, and world models, with a focus on spatiotemporal modeling, 4D dynamic reasoning, long-horizon prediction, and physical consistency. Further interests include unified understanding and generation, and perception and interaction for embodied intelligence.

Education

Sun Yat-sen University Project 985 / 211 / Double First-Class
B.Eng. in Intelligent Science and Technology

SEP 2023 -- JUN 2027
GPA 3.91 / 4.0 Major GPA 3.94 / 4.0

Research Experience & Publications

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Chaoyue Li*, **Yongxue Xu***, Jie Feng*, and Jiayu Ding†

Submitted to NeurIPS 2026; [arXiv:2605.19390](https://arxiv.org/abs/2605.19390)

Developed [LMM-Track4D](#) for trajectory-grounded multi-turn dialogue: answering spatiotemporal questions about multi-view videos while predicting query-conditioned 3D object trajectories. Combined geometry-aware encoding, persistent state memory, and structured trajectory decoding; co-developed Track4D-Bench.

[Equipped LMMs with 4D dynamic reasoning capabilities. Substantially outperformed the base model on Track4D-Bench and external benchmarks, especially in dynamics reasoning.](#)

Contribution: Led model design and main experiments; co-developed the benchmark.

Learning to Reason with Persistent Object States for Video Instance Segmentation

Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, Rui Qian, and Linfeng Zhang†

Submitted to ICLR 2027

Developed [POSReasoner](#) for video instance and object segmentation, preserving identities as objects become occluded and reappear. Coupled persistent object memory with verified state transitions to prevent ambiguous observations from corrupting identity history.

[Improved long-horizon identity continuity and object recovery with the segmentation models frozen. Delivered consistent gains across evaluated backbones and benchmarks, including occlusion-heavy and long-video settings.](#)

Contribution: Led method and architecture design, implementation, experiments, and manuscript preparation.

TRACE: Typed Reference-Aware Conversational Editing for Humanoid Motion Diffusion

Zejian Kang, Yifu Luo, Boyang Shen, Jingxuan Li, Zekai Chen, Shi Zekun, **Yongxue Xu**, Zhenyu Hou, Bingqian Wu, Shaolei Zhang, Wentao Yang, and Xiangru Huang

Submitted to ICLR 2027

Contributed to TRACE for multi-turn, language-guided humanoid motion generation and editing. Typed motion programs specify what to change or preserve, while reference-constrained diffusion modifies selected joints and time intervals without regenerating the entire motion.

Contribution: Reproduced baseline methods, adapted evaluators, and converted motion demos for Unitree G1.

Latent State Tracking for Persistent Object-Centric Representations in Video Generation

Ongoing Research

Developing an object-centric latent dynamics framework for language-guided video generation, coupling event-conditioned state prediction with observation-based state correction. Predictive state trajectories condition generation, aiming to preserve semantic state changes and object identity over long horizons, including periods without direct visual observation.

Graph Domain Adaptation Does Not End with Representation Learning

Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang†

Submitted to ICLR 2027

Developed [EviGDA](#) for unsupervised graph domain adaptation: transferring node classifiers from labeled source graphs to unlabeled target graphs by fusing graph-aware and feature-only predictions.

[Surpassed existing state-of-the-art methods in overall graph domain adaptation performance.](#)

Contribution: Designed the core method and led implementation, evaluation, and manuscript development.

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang†

Under review at IEEE Transactions on Network Science and Engineering

Developed [DRCIM-ML](#) for robust seed selection in competitive multilayer networks under node-removal attacks, using diffusion-aware role alignment and cross-layer evolutionary search.

[Extended RCIM from single-layer to multilayer networks, with a new formulation and a dedicated solver.](#)

Contribution: Led problem formulation, method design, experiments, and writing; one patent pending.

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang†

GBCESC 2025 / [Best Paper Award](#)

Studied critical-node identification in urban road networks through comparative evaluation of graph neural network and deep reinforcement learning approaches.

Contribution: Dataset construction, GNN/DQN baseline reproduction, and experimental visualization.

* Equal contribution. † Corresponding author.

Research Tools

[AstraDraw](#). Developed a reference-guided workflow for editable PowerPoint research figures, reusable visual assets, and iterative local refinement. Used in my own papers; attracted [100+ GitHub stars](#).

[CCFA-Skills](#). Co-developed with [InkMind.AI collaborators](#): a coordinated toolkit of 17 research skills spanning literature, experiments, writing, figures, review, and submission. Integrated AstraDraw for reference-guided figure authoring.

Research Initiatives & Collaborations

Collaborate with researchers at [HKUST](#), [SJTU](#), and [Westlake University](#) on [video generation and multimodal spatiotemporal understanding](#); explore [AI-assisted research](#) with peers in [InkMind.AI](#).

Awards & Competitions

Sun Yat-sen University First-Class Scholarship
2023–2026

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize
Dec. 2024

Mathematical Contest in Modeling (MCM), USA – Honorable Mention
Feb. 2025

National Undergraduate Smart Car Competition, South China Region – Second Prize
Aug. 2025

Beyond Research: Table tennis, music, and nature travel.